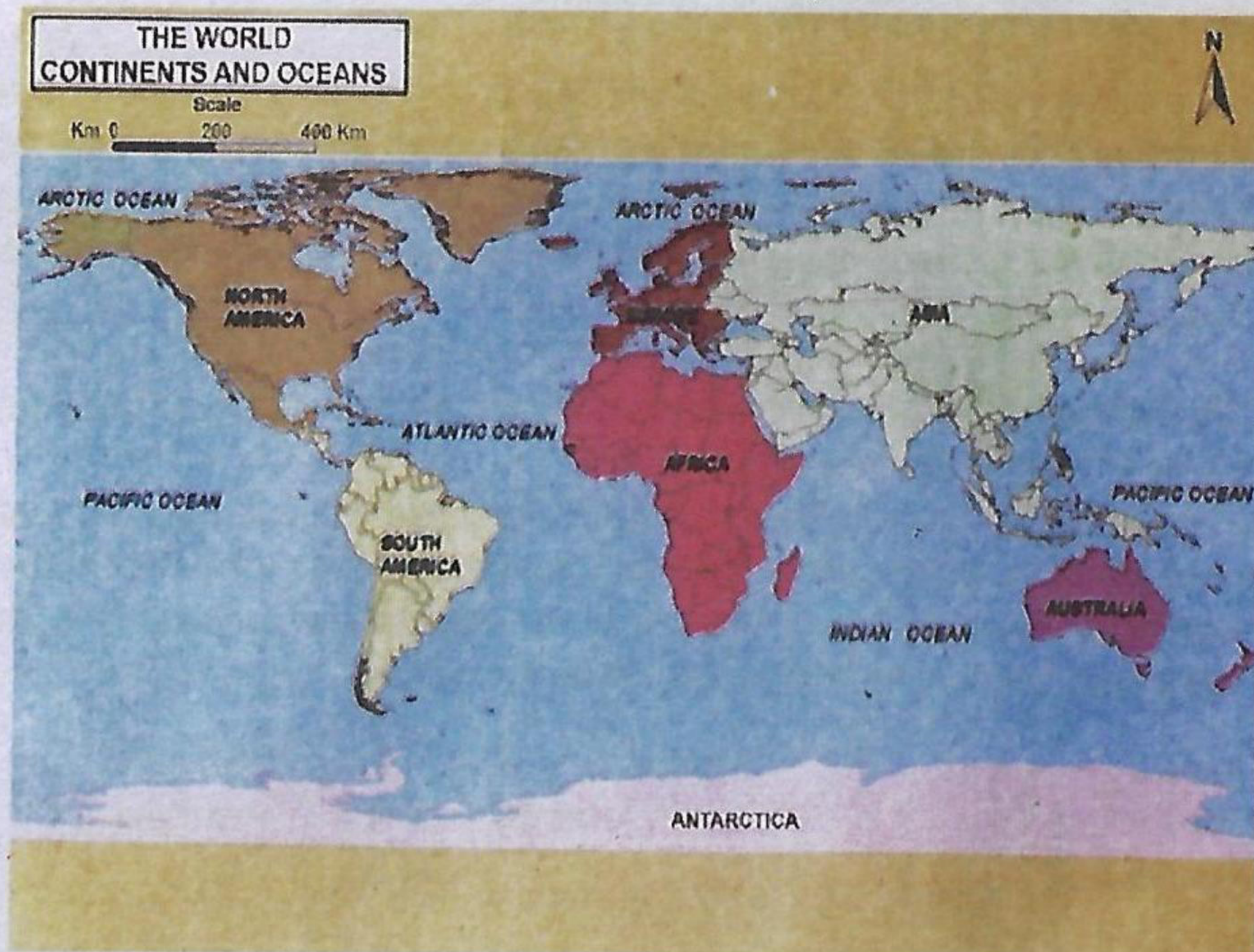
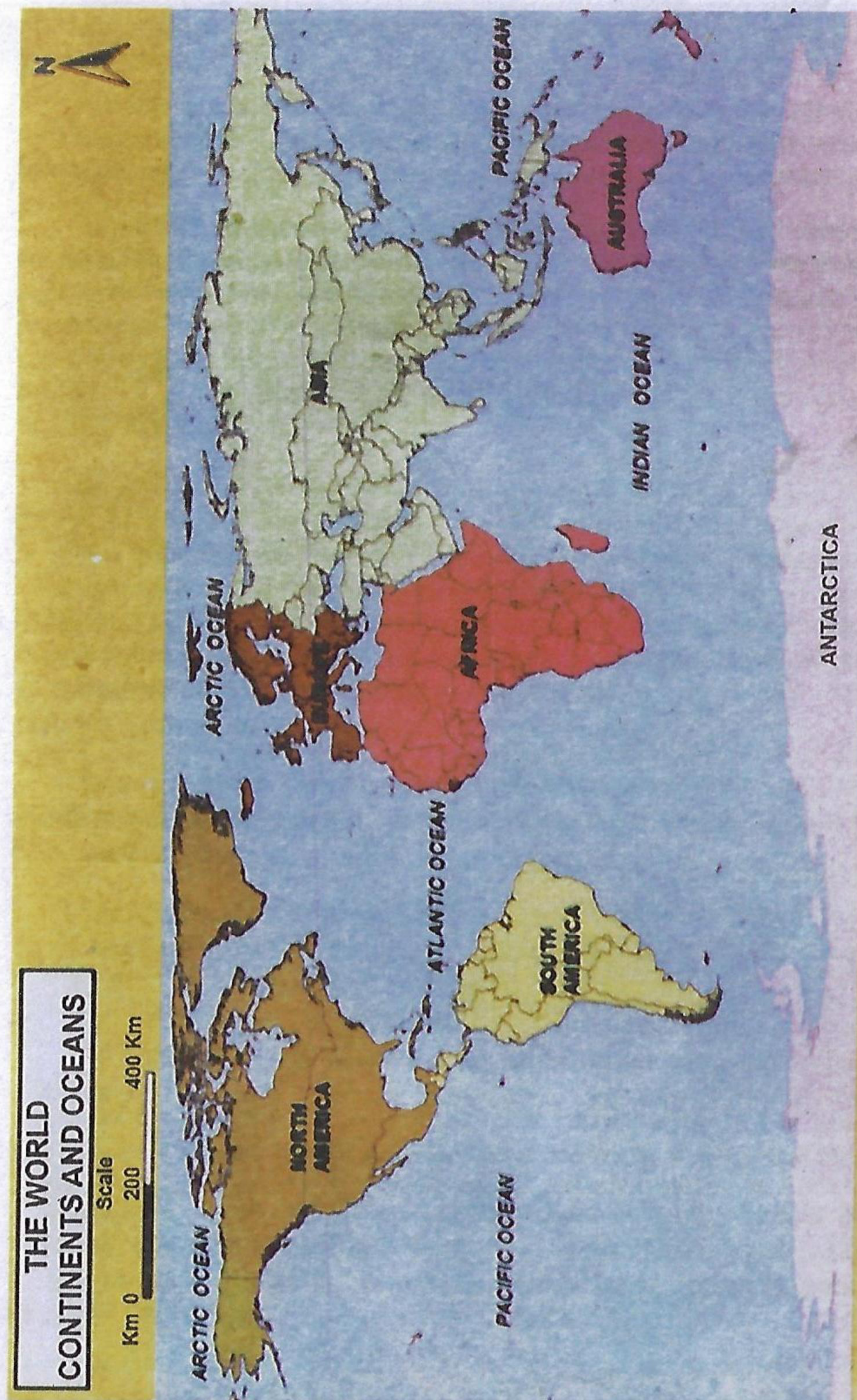


Several plateaus lie between the mountain ranges of Central Asia. The highest is the Tibetan Plateau, often referred to as the Roof of the World, which is bounded by the Kunlun Mountains and the Himalayas.



In Asia flow some of the important rivers of the world, among them Amu, Huang; the Indus, Ganges, and Yang Se Kiang are worth mentioning. The major rivers of South Asia have their sources in the Himalayas. The Ganges rises in the western Himalayas and passes eastward through India. Just north of the Bay of Bengal, it joins the Brahmaputra river, which rises beyond the Himalayas and then empties into the bay. The Indus river emerges from the western end of the Himalayas and flows through Jammu and Kashmir and Pakistan to the Arabian Sea. The only large rivers of Southwest Asia are the Tigris and the Euphrates. Both rivers rise in Turkey and flow southward through Syria into Iraq, where they join before emptying into the Persian Gulf. Lake Baykal in southeastern Siberia is the deepest lake in the world and the largest freshwater lake in Asia.

Most of Asia's climates are similar to the interior and eastern-coast climates of North America at similar latitudes. Like northern Canada, the northernmost areas of Asia have a sub-polar climate with very long, cold winters and very short, cool summers. Vast area with a sub-arctic climate lies farther inland and generally southward. This area experiences great extremes of temperature. South of the sub-arctic regions is a broad stretch of land having a humid continental climate with



Europe

Europe conventionally is one of the seven continents of the world. Although referred to as a continent, Europe is actually just the western fifth of the Eurasian landmass, which is made up primarily of Asia. Modern geographers generally describe the Ural Mountains, the Ural River, part of the Caspian Sea, and the Caucasus Mountains as forming the main boundary between Europe and Asia. The name Europe is perhaps derived from Europe, the daughter of Phoenix in Egyptian mythology, or possibly from Ereb, a Phoenician word for "sunset."

Europe is the second smallest continent (Australia is the smallest) in the world, which has an area of 10,355,000 sq. kilometres, but it has the third largest population of all the continents i.e. 741 million in 2016.

Europe is the most developed continent of the world. In its east is Asia, in the west is Atlantic Ocean, in the North is Antarctic Ocean and in the South is Mediterranean Sea. According to area this continent stands sixth among the various continents. Europe has long been a centre of great cultural, economic and scientific achievement. The ancient Greeks and Romans produced major civilizations, famous for their contributions to philosophy, literature, fine art, and government.

Some important countries of Europe are Germany, Britain, Norway, Sweden, Poland, Spain, France, Denmark, Switzerland, Holland, Italy, Spain, Greece, Russia, Ukraine and Byelorussia, etc.



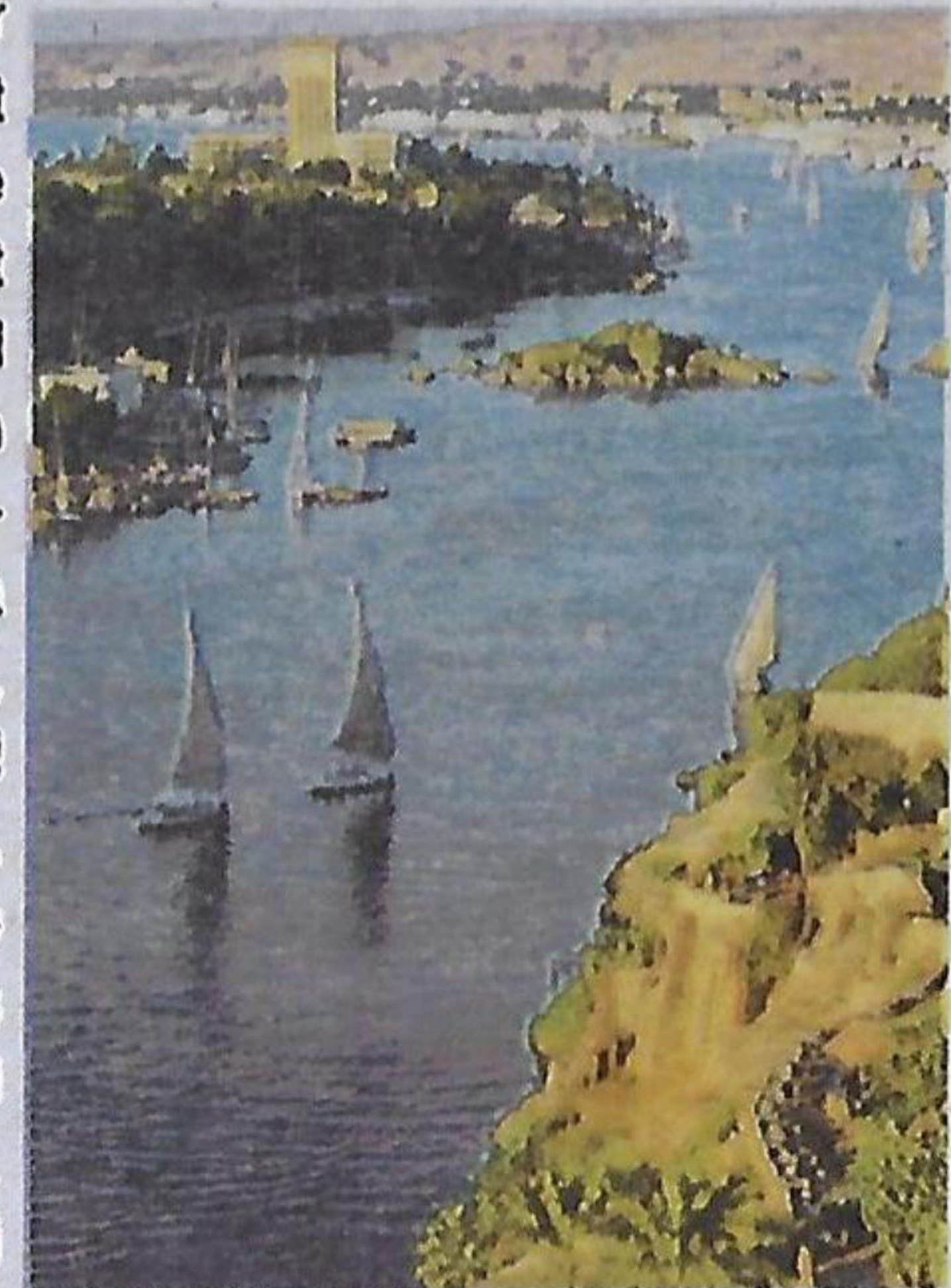
London Bridge

Due to many significant features Europe has great importance amongst the rest of the continents. This includes the temperate climate and the coastal areas of Europe. Due to moderate climate people remain fit and energetic the whole year round. The coastal area of Europe is most suitable for good seaports. Some best seaports of the world are found in Europe, for example, London, Hamburg, Copenhagen, Rome, etc. Moscow, St. Petersburg, and Paris are important cities. The most important mountain range of Europe is Mount Alps. Europe is also rich in natural resources and has highly developed means of transport, agriculture and industry.

Africa

Africa is the second largest of the Earth's seven continents, covering an area of 30,244,000 sq. kilometres including its adjacent islands. It comprises 23 per cent of the world's total land area. In the year 2016 A.D. some 13 per cent of the world's population, an estimated 1.216 billion people, lived in Africa, making it the world's second most populous continent, after Asia.

In the east of Africa is Asia and Indian Ocean. In the west is Atlantic Ocean and in the North is Mediterranean Sea and in the South is Indian Ocean. Other than Northern Atlas Mountain there are no other significant mountain ranges. In Africa the biggest desert of the world is found which is called the Sahara Desert and the River Nile is an important river of this continent. There are not many good seaports. Most of Africa's population lives in the region, south of the Sahara, known as sub-Saharan Africa. This area called eastern Africa includes countries such as Ethiopia, Somalia, and Uganda. Among the nations of Central and West Africa are Angola, Cameroon, Ghana, Nigeria, and Democratic Republic of the Congo (DRC, formerly Zaire). Southern Africa is dominated by the country of South Africa and also includes Botswana, Lesotho, and Namibia. Since 1950s, most African nations have gained independence from their former colonial powers.



River Nile

African coastlines are not similar to the coastal areas of Europe. Alexandria, Port Saeed and Cape Town are most important seaports. Cairo, Tripoli, Khartoum and Nairobi are important cities. With the world's highest population growth rate, Africa continues to face challenges in its industrial and social development.

North America

According to area this continent is the third largest continent of the world. In its east is Atlantic Ocean and in its west is Pacific Ocean and in the north is Arctic Ocean and in the south is South America. North America includes Canada (the 2nd largest country in area in the world), the United States (3rd largest), and Mexico (14th largest). The continent also includes Greenland, the largest island, as well as the small French overseas department of Saint-Pierre and Miquelon and the British dependency of Bermuda both made up of small islands in the Atlantic Ocean. The name America is derived from that of Italian navigator Amerigo Vespucci, who may have visited the mainland of North America between 1497 and 1498. The area of the continent is 23.7 million sq. kilometres.

The continent has three enormous coastal indentations: Hudson Bay in the northeast, the Gulf of Mexico in the southeast, and the Gulf of Alaska in the northwest.

North America can be divided into five major natural regions. The eastern half of Canada, as well as most of Greenland and sections of Minnesota, Wisconsin, Michigan, and New York in the United States, are part of the Canadian Shield (or Laurentian Plateau). The second region is made up of a coastal plain in most of the eastern United States and Mexico. In the United States the coastal plain is bordered on the west by a third region, comprising a relatively narrow cordillera of mountains and hills, notably the rounded Appalachian Mountains. A fourth region consists of the central portion of the continent, from southern Canada to southwestern Texas. The fifth, and the westernmost region of North America, taking in most of Mexico, is an active zone of mountain building; its recent geological history is dominated by the volcanic activity.

In 2002 A.D., the United States had 280,562,490 inhabitants; Mexico 103,400,170; Canada 31,902,268; and Greenland 56,376.

The two countries of North America, the USA and Canada are the highly developed nations, whereas Mexico is an underdeveloped country. The famous mountain range of North America is Rocky Mountains and the most famous river of this continent is River Mississippi. This continent was discovered and made habitable by the Europeans among them, the Spanish, the British and the French are the most significant. These nations utilized the natural resources of Northern America very well and have become the most developed nations of the world.

The largest cities of this continent include Mexico City, Guadalajara, and Monterrey, in Mexico; New York City, Los Angeles, Chicago, Houston, Philadelphia, and San Diego, in the United States; and Toronto, Montreal, Vancouver, Ottawa, and Edmonton, in Canada.

South America

South America, the fourth largest of the Earth's seven continents (after Asia, Africa, and North America), occupying 17,820,900 sq. kilometres or 12 per cent of the Earth's land surface. It lies astride the equator and tropic of Capricorn and is joined by the Isthmus of Panama, on the north, to the

Central and North America. In the east of South America is Atlantic Ocean in the west is Pacific Ocean, and in the south is Antarctic Ocean.

South America has an estimated population of 348 million, or 6 cent of the world's people. The continent comprises 12 nations; ten Latin states are Argentina, Bolivia, Brazil, Chile, Colombia, Ecuador, Paraguay, Peru, Uruguay, and Venezuela and two of the nations are former dependencies: Guyana, of the United Kingdom, and Suriname, of the Netherlands. South America also includes French Guiana, an overseas department of France.

The majority of the people speak Spanish language in this region. The countries of this continent are less developed as compared to the countries of North America.

In South America, the famous mountain range is called Mount Andes, which is the second largest mountain range in the world. The main lowland is the vast



Amazon Basin in the equatorial part of the continent, which is drained by the Amazon River, the world's second longest river.

South America is dominated by relatively warm climatic regions.

Some important cities of this continent are Rio de Janeiro, Buenos Aires, Brasilia and Sao Paulo, etc. South America's overall population has been increasing rapidly, especially in the developing tropical countries, and urban populations have increased greatly in all parts of the continent. The overall population density is 20 persons per sq. kilometre but more than half of the continent has a population density of fewer than 2 persons per sq. km. The total population of this continent in 1998 was 331 million.

Spanish is the official language of nine of the thirteen countries of this continent. About 90 per cent of the total population is Roman Catholic.

Australia

Australia is an island continent, which is located in the southeast of Asia, forming with the nearby island of Tasmania, the Commonwealth of Australia, a self-governing member of the Commonwealth of Nations. The continent is bounded on the north by the Timor Sea, the Arafura Sea, and the Torres Strait; on the east by the Coral Sea and the Tasman Sea; on the south by the Bass Strait and the Indian Ocean; and on the west by the Indian Ocean.

Australia is the smallest continent of the world. The total population of this continent in 1998 was 18.7 million.

The Commonwealth of Australia is made up of six states—New South Wales, Queens land, South Australia, Tasmania, Victoria, and Western Australia—and two territories—the Australian Capital Territory and the Northern Territory.

The first people who lived in Australia, called Aborigines, migrated there about 40,000 years ago. Many Europeans specially the British have settled in this continent. Australia comprises two big countries (Australia and New Zealand) and many islands. The larger portion of Australia is a desert and the habitation is limited to the coastal areas only.

The climate of Australia differs from region to region but has no extreme climate anywhere. The eastern, western and southern coastal regions are the densely populated areas. This continent is very famous for rearing cattle. The famous cities of this continent are Sydney, Perth, Adelaide, Melbourne and Wellington etc.

As this continent is densely populated, therefore it is entrapped in the problems of dirt, illiteracy and hungry. It's environment is very fascinating. People enjoy a good health. They have all the facilities of life. They spend a comfortable life.

Antarctica

Antarctica is the fifth largest of the Earth's seven continents. With an area of 14 million sq. km. Antarctica is larger than either Europe or Australia.

This is an uninhabited continent where snow falls throughout the year. The land is covered by snow several meters deep. No one can live here permanently. The scientists of different countries come and stay here only for scientific research and experiments. Scientists coming to this part of the world mostly belong to the developed nations, for example, America, Britain, France, Russia and Australia. Pakistan has also established a research station in this region. Seven nations—Argentina, Australia, Britain, Chile, France, New Zealand, and Norway—claim territorial rights in Antarctica. The total volume of the ice sheet covering Antarctica is estimated to be about 90 per cent of the world's ice.

Oceans

Like land, water is also divided into different divisions. Even though Oceans are all joined together but they have been named after the areas they surround. There are five Oceans: Pacific Ocean, Atlantic Ocean, Indian Ocean, Arctic Ocean and Antarctic Ocean. These oceans are further divided into seas, lakes, bays, gulfs. For example, Mediterranean Sea, Arabian Sea, China Sea, Bay of Bengal, Gulf of Mexico and Strait of Malacca etc.

Pacific Ocean

This is the largest ocean of the world. It covers more than a third of the Earth's surface and contains more than half of its free water. It is sometimes divided into two nominal sections: the part north of the equator is called the North Pacific and the part south of the equator is called the South Pacific. The name Pacific, which means peaceful, was given to it by the Portuguese navigator, Ferdinand Magellan in 1520.

In the east of Pacific Ocean are North and South America, in the west is Asia and Australia, in the north is Arctic Ocean and in the south is Antarctic Ocean. The Pacific is the oldest of the existing ocean basins, its oldest rocks dating back to 200 million years. The Pacific Ocean contains more than 30,000 islands; their total land area, however, amounts to only one-quarter of one per cent of the ocean's surface area.

Pacific Ocean is spread over 25 per cent of the total area of the Earth. It has many seas, for example, Sea of Japan, South China Sea, East China Sea, etc. The depth of the ocean is approximately 4 km. The deepest point is near Philippines. Along the coastal areas of this ocean, mountain ranges are spread over a large area, which are very famous for Earthquakes. Pacific Ocean has many big and small islands including the Island of Philippine, Japan and Hawaii etc. This ocean is not very significant for international trade because there are no good natural harbours.

Atlantic Ocean

Atlantic Ocean, the second largest of the Earth's five oceans is the most heavily travelled ocean on the Earth. Only the Pacific Ocean is larger, covering about twice the area of the Atlantic Ocean. The Atlantic is divided into two nominal sections: The part north of the equator is called the North Atlantic; the part south of the equator, the South Atlantic. The ocean's name is derived from Atlas, one of the Titans of Greek mythology. The continents of Europe and Africa are on its eastern side and the continents of North America and South America on its Western side. In the North is Arctic Ocean and in the South is Antarctic Ocean. The depth of this ocean is approximately 3-¼ km.

Atlantic Ocean is a significant sea highway as the world's trade takes place through this ocean. The main reason for this is that in its east are the developed nations of Europe and in the west are developed countries of North and South America from where various commodities are exported to various parts of the world and different items are imported from different countries. The Atlantic Ocean contains some of the world's most productive fisheries, located on the continental shelves and marine ridges off the British Isles, Iceland, Canada (especially the Grand Banks off Newfoundland), and the northeastern United States.

In this ocean, many big and small islands are found. Among them the islands of Cuba and West Indies are most significant. In the seas of Atlantic Ocean, Caribbean Sea, Baltic Sea and Mediterranean Sea are quite famous. In bays, Hudson Bay and Gulf of Mexico are famous.

Indian Ocean

Indian Ocean is the third largest of the Earth's five oceans, bounded on the west by Africa, on the north by Asia, on the east by Australia and the Australasian islands, and on the south by Antarctica. No natural boundary separates the Indian Ocean from the Atlantic Ocean, but a line 4,000 km. long on the 20th meridian east of Greenwich, connecting Cape Agulhas at the southern end of Africa with Antarctica, is generally considered to be the boundary between the two oceans.

The Indian Ocean contains numerous islands, the largest of which are Madagascar and Sri Lanka. Smaller islands include the Maldivian group and Mauritius.

The Arabian Sea, the Persian Gulf, the Gulf of Martaban and the Bay of Bengal are all offshoots of the Indian Ocean.

In the coastal areas of the Indian Ocean, 2/3rd of the world's population is settled which has increased its trade significance.

Indian Ocean is an ancient sea route. International ships after passing through the Suez canal enter this ocean. The ships coming from South Africa also enter this ocean. Some important seaports of Indian Ocean are Karachi (Pakistan), Aden (Yemen), Nairobi (Kenya), Dares Salaam (Tanzania), Mumbai, Chennai, Suranga Patum (India), Colombo (Sri Lanka), Yangon (Myanmar) and Bangkok (Thailand).

Arctic Ocean

The Arctic Ocean is the smallest of the world oceans. This Ocean extends South from the North Pole to the shores of Europe, Asia, and North America.

The surface waters of the Arctic Ocean mingle with those of the Pacific Ocean. The major portion of this ocean remains frozen. Therefore, it does not have much navigational value. There are very few seaports because its coastline is uninhabited and remains unfit for shipping. This ocean along with Atlantic Ocean and Pacific Ocean through the eastern and western straits of Greenland is joined with Bering Strait.

Three forms of ice are found in the Arctic Ocean: Land ice, river ice, and sea ice. Land ice enters the ocean in the form of icebergs, which are created when pieces of glaciers break off.

Antarctic Ocean

This ocean surrounds the continent of Antarctica. It is joined to Pacific Ocean, Atlantic Ocean and Indian Ocean in the south. This ocean also has no navigational importance, as it remains frozen throughout the year. Economically, this ocean does not have much importance.

Table showing the average volume, depth and area of the World Oceans

Ocean Or Sea	Average m	Depth ft	Area million sq km	Volume million sq mi	million cu km	million cu mi
Pacific Ocean	4,300	14,000	165.7	64.0	707.6	169.9
Atlantic Ocean	3,900	12,900	82.4	31.8	324.6	77.9
Indian Ocean	3,900	12,800	73.4	28.3	291	69.9
Arctic Ocean	1,300	4,300	14.1	5.4	17	4.1
Australasian Central Sea	1,200	3,900	8.1	3.1	9.9	2.4
Gulf of Mexico and Caribbean Sea	2,200	7,300	4.3	1.7	9.6	2.3
Mediterranean and Black Seas	1,430	4,690	3.0	1.1	4.2	1
Bering Sea	1,440	4,720	2.3	0.9	3.33	0.8
Sea of Okhotsk	838	2,750	1.6	0.6	1.3	0.3
Hudson Bay	101	331	1.2	0.5	0.16	0.04
North Sea	94	310	0.6	0.2	0.05	0.01
Baltic Sea	55	180	0.4	0.2	0.02	0.005
Irish Sea	60	200	0.10	0.04	0.006	0.001
English Channel	54	180	0.08	0.03	0.004	0.001
The World Ocean	3,790	12,430	361.1	139.4	1,370	329

Sea Movements

The hydrosphere never remains stagnant rather it is always in motion. These movements are of three types.

1. Waves
2. Tides
3. Currents.

Waves

When wind blows the level of water starts rising up and down which is called a wave. in a pond where waves are rising if a small piece of plastic is thrown, it will be observed that waves keep on moving continuously but the plastic does not travel with the waves from one point of the pond to the other. This shows that if waves are seen moving from one place to another the water does not flow with them.

At various times gigantic waves strike the surrounding shore with tremendous force and cause considerable damage to life and property. Although sometimes called tidal waves, these waves are not caused by tidal phenomena. Earthquakes, undersea volcanic eruptions, and underwater landslides can cause large waves to wash ashore that are known as Sunamis. Hurricanes can cause a dome of water to wash ashore that is known as a storm surge.

During the movement of waves the water drops keep rising up and down and with their movement also make the other drops movable. High and low winds cause high and low waves. In open sea the average height of the waves could be up to 15 metres. The highest point of the waves is known as 'peak' and the lowest is known as 'trough' When the waves reach the shore from the centre of the sea then their pressure breaks after striking with the banks.

Tides

People living in coastal regions are well aware of the tides. Tides are basically the rise and fall of seawater. Tides are caused due to the force of gravity of Sun and moon. As compared to the Sun the effect of the gravitational force of the moon is more because the moon is closer to the Earth than the Sun. The distance between the moon and the Earth is 385,000 km. whereas the distance between the Sun and the Earth is 150 million. With the movement of the moon, the height of the tide also increases. Because the force of gravity decreases with distance, the moon exerts a stronger gravitational pull on the side of the Earth that is closer to it and a weaker pull on the side farther from it. The Earth does not respond to this variation in strength because the planet is rigid—instead, it moves in response to the average of the moon's gravitational attraction. The world's oceans, however, are liquid and can flow in response to the variation in the moon's pull. On the side of the Earth facing the moon, the moon's stronger pull makes water flow toward it, causing a dome of water to rise on the Earth's surface directly below the moon. The dome of water directly beneath the moon is called direct tide, and the dome of water on the opposite side of the Earth is called opposite tide.

The Sun also gives rise to two oppositely situated domes, but because the Sun is far away from the Earth, its tide-raising force is only about 46 per cent to that of the moon.

There are several advantages of the tides, for example, in those seaports where water is deep, big ships can get anchored easily due to the tides, which increase the depth of the water. The tides also bring with them precious shells to the shore.

Currents

Apart from the waves and the tides when the water of the sea starts flowing from one place to another in the form of a river then it is called an ocean current. A current flows constantly on a definite route like a river and it has its own width. The difference between a river and an ocean current is that a river flows on land whereas an ocean current flows in sea. The other difference is that the width of the river is small and the depth of the water is low, whereas the width of ocean current is very big and the depth is also very high. The width of the ocean current is maximum 500 km. and its depth is approximately 180 metres. The average speed of ocean current is generally 16 km. per hour.

The Reasons for the Formation of Ocean Currents

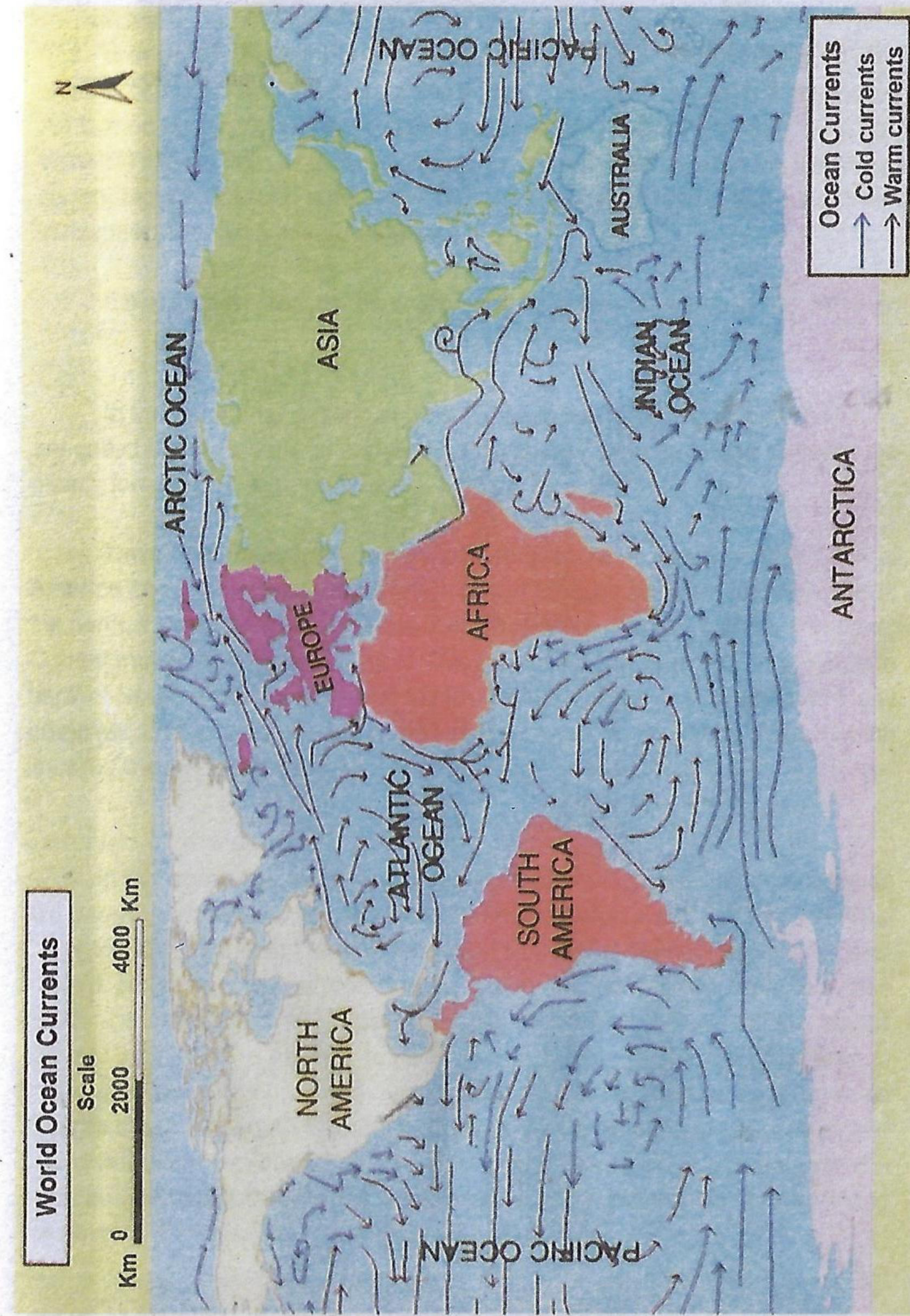
Just as there are reasons for the formation of waves and tides, similarly, there are many reasons for the formation of currents also. A few are discussed here.

1. The difference of Temperature

In the equatorial regions, the rays of the Sun fall vertically therefore these regions have hot seawater. Since hot water is lighter therefore the level of the sea is comparatively high. In contrast to this, in Polar Regions the rays fall obliquely, which is less capable of heating the seawater. Therefore, the seas of these regions are extremely cold. Cold water is heavier and starts settling down, creating a slope between equatorial regions and Polar Regions due to which ocean currents are formed.

2. The High and Low Quantity of Salt

The low quantity of salt in water is also due to temperature. In all those seas where due to high temperature the process of evaporation is high the quantity of salt increases. Due to this water becomes heavier and gets settled down and starts flowing from down the sea in the form of a current to the seas with less salty water. Less salty water is lighter and fresh and therefore, stays less near the sea level. For example, in the Mediterranean Sea the process of evaporation is high due to which the water is more salty there. Contrary to this, in the waters of Atlantic Ocean the process of evaporation is slow. Therefore, this water is fresh and light. As a result, a series of ocean current starts flowing between Atlantic Ocean and Mediterranean Sea.



3. The Orbital Rotation of the Earth

The Earth completes its rotation on its axis in 24 hours. Due to this rotation a change occurs in the direction of the ocean currents. The ocean currents are not formed due to the Earth's rotation but according to the Feral law, the ocean current is moved from its actual direction. In the northern hemisphere, the ocean current moves from the actual direction towards the right and in the southern hemisphere it moves towards the left.

Each ocean has its own process of currents which is as follows:

Currents of the Pacific Ocean

In the equatorial regions of Pacific Ocean two big currents emerge which are called North Equatorial and South Equatorial Currents. These currents are mainly formed due to trade winds.

The North Equatorial Current moves from the western bank of Central America towards the west up to the islands of Philippines. Then turning towards the north, it enters the Sea of Japan. Here, it is called Kuroshio. South Equatorial Current moves from the North West. This current starting from the North Western bank of South America moves up to New Zealand. Then it is divided into many branches. One of its branches again join the flow of South Pacific Ocean which is called the Antarctic Current.

Currents of the Atlantic Ocean

In this ocean too, in the north and south of the Equator due to trade winds two ocean currents are formed which are called North and South Equatorial Currents of the Atlantic Ocean. The North Equatorial Current flows from east to west. Near the island of West Indies it gets divided into two branches. One moves towards the Gulf of Mexico and the other towards the north of the islands of West Indies to the eastern bank of America. Here it joins its other branch and in the form of a combined current flows parallel to the coasts of America towards the north. South Equatorial Current also moves towards the west and gets divided into two branches near Brazil. One branch flows parallel from the north coasts of South America to the west and the other branch flows along the eastern coasts towards the south.

The Currents of the Indian Ocean

The currents of the Indian Ocean are slightly different from the currents of the Pacific Ocean. Monsoon winds blow in the Indian Ocean, which do not have the same direction throughout the year; rather they keep on changing their directions in all the seasons. Along with them the ocean currents also change their direction. Therefore, during the winter season in the Northern Hemisphere, the monsoon winds blow from Asia (South Asia) towards the Indian Ocean from the Strait of Malacca a current flows which moves along the coasts of Malaysia, Myanmar, Bangladesh, India, Sri Lanka, Pakistan and East Africa. On reaching Pakistan, the direction of this wind is moved towards the west by the northeastern monsoon winds.

The Effects of Ocean Currents

The ocean currents affect the climate, seaports, navigation and fishing. Some of their effects are mentioned below:-

1. The wind, which blows over a warm current, also gets warm due to which the capability of absorbing the water vapour rises in the winds. When these winds rise up, they get cold and due to the process of condensation result in rainfall.
2. Apart from rainfall the ocean current also keeps the temperature of coastal regions moderate. The temperature of those cold countries becomes moderate where the warm currents flow along the coast. For example, had the equatorial warm current not passed near Britain, its seaports would have frozen and no trade could have taken place. Due to the effect of these ocean currents these seaports remain open throughout the year. These currents, thus have a positive impact on the economy of Britain. Contrary to this, near the coast of Siberia cold currents flow due to which this coast remains frozen for ten months in a year and there is no important seaport here.
3. The point where warm and cold currents meet, cause the winds blowing over them to strike against each other, due to which storms occur. These types of storms are known as Hurricanes in the US and Typhoons in Japan.
4. Even though in the present times, due to the invention of steam and atomic powers the seaships do not require the help of currents but still navigators prefer moving along the direction of these currents and save time and energy.

5. The fish of the warm areas reach the cold areas and vice versa through these currents only. In this way, the quantity of fish does not decrease around the world.
6. The point where warm and cold currents meet, fish is found in abundance. For example, near America Newfoundland (Canadian province), the coastal area of Iceland and Norway in Northern Europe are also famous for fish.

Water is very essential for life. Without it we can hardly stay alive for a few days. Only Clean water, free from germs should be used. Our villages and cities are supplied water from rivers, lakes and tubewells. With the development of industries, generally the factories are installed near rivers. The Industrial waste containing poisonous chemicals are disposed off in the rivers which is extremely harmful for human beings and marine life. From these poisonous chemicals not only the marine life gets affected but also human beings using this contaminated water, are in danger of contracting several harmful diseases. The factories near the coast of Karachi are also a big threat to the marine life, because of the chemical waste that they are depositing in the seas. Similarly, the oil released in sea from the ships, is also proving dangerous for the marine life. Many species of animals in the sea are in constant danger. Government of Pakistan is trying to convince the factory owners to adopt effective measures and not to throw the poisonous chemical waste into the rivers and seas. It is our duty to help the government in this work for our own survival and for the survival of the future generations.

ENVIRONMENTAL POLLUTION

Definition of Pollution

Pollution is the contamination of the Earth's environment with materials that are dangerous for human health, the quality of life, or the natural functioning of ecosystems (living organisms and their physical surroundings).

There are many factors that pollute our environment and we ourselves are responsible for them. For example, gutter water, heaps of garbage, the chemicals from factories, the gases coming out from the factories and motor cars and smoke, growing use of chemical fertilizers, insecticides, etc.

Water is the best solvent and dissolves many things. Many gasses, liquids and solids dissolve easily in water. In this way, it loses its real form and gets polluted. All the things dissolved in water bring about some changes in it. If harmful mixtures get into water then this is called water pollution. Due to the rapid increase in pollution the quality of water is also changing. The presence of dissolved oxygen in the river Indus is 2.3 milligrams per litre which is less than the average ratio of 5.5 milligrams per litre. Contaminated water is not only harmful for human health but also leaves negative effect on the animals and plants as well.

Soil Pollution

The presence of harmful elements in the soil makes it polluted biologically, chemically and physically. In the present world like air and water, soil is also exposed to pollution. Apart from the pollutants that are directly polluting it, air and water pollution is also destroying the soil.

Soil also gets polluted by the natural and artificial elements. For example, the fire in the forests, floods, earthquakes and the eruption of the volcanic mountains pollute the soil naturally. But human activities are the main reason of soil pollution. The process of deposition of domestic, agricultural and industrial waste in the soil is being carried out since long, which not only affects the surface of the Earth but also affects the soil water and its fertility.



Soil Pollution

The waste deposited in the soil includes agricultural waste, rubbish, waste paper, waste of metals, empty boxes, rubber, plastic, leather, cloth, the useless items of ceramics, left over food, the waste of slaughter houses, construction waste, dead animals, chemical waste, insecticides in large quantities and the industrial waste.

The taking away of important elements, which help in making the soil fertile and also help in the growth of vegetation in the soil by the rainfall and snow is also a kind of pollution. Many factors make the agricultural soil barren and unfit for cultivation.

A few factors are mentioned below:

1. Due to deforestation and rearing of the cattle the soil becomes barren.
2. The growth of salinity and water logging, due to the use of polluted water for irrigation.
3. The deposition of poisonous chemical wastes, due to the use of chemically made medicines.
4. The excessive use of chemical fertilizers.
5. The collection of harmful industrial waste in the soil.
6. The effects of air and water pollution.

All the harmful elements polluting water and air eventually pollute the soil and enter into the human body, animals and plants by the process of food chain.

Exercise

A. Answer the following questions:

1. What is the ratio of water and land on Earth?
2. Into how many continents is the World divided?
3. Name at least five countries of Asia.
4. State some of the reasons for the development of Europe.
5. Which ocean has the greatest depth?
6. What is meant by waves?
7. How are tides formed?
8. What are the ocean currents?
9. State the reasons for the formation of currents.

10. How are ocean currents helpful to navigators?
11. What is meant by pollution? State its kinds.

B. Fill in the blanks:

1. According to _____ Asia is the biggest continent of the world.
2. _____ is the biggest ocean of the world.
3. Permanent winds are also called _____.
4. _____ is the rising and falling of sea water.

C. There are three possible answers given for the following questions. Tick✓the correct answer.

1. As compared to the Sun, the moon's gravitational force affects the Earth more because _____.
 1. The Moon is closer to the Earth than the Sun
 2. The Sun is closer to the Earth than the moon
 3. The Earth is closer to the Sun than the moon
2. According to World Health Organization every adult individual requires almost _____ kilograms of air.
 - a) 20 kilogram b) 15 kilogram c) 40 kilogram
3. At present the standard of living is measured on the basis of:
 - a) Agricultural development b) Industrial development
 - c) Mineral development.
4. Which is the most commonly spoken language in Asia:
 - a) Chinese b) Japanese
 - c) English
5. The biggest desert of the world is found in _____.
 - a) Africa b) Asia. c) Europe.

6. The third largest continent of the world is _____.

a) Australia b) North America c) Antarctica

7. Mount Andes is the _____ mountain range in the world.

a) Largest b) Second largest c) Third largest

8. The name Pacific means _____.

a) Peaceful b) Pure c) Parity

Activities

1. Show the location of at least five countries of each continent of the world on the map.
2. Make a list of those features, which pollute our environment.
3. Find out at least ten effects of pollution on the environment.
4. Celebrate 'Earth Day' on 22nd April with special focus on factors causing pollution. Make a presentation in the School Assembly to create awareness among the students.